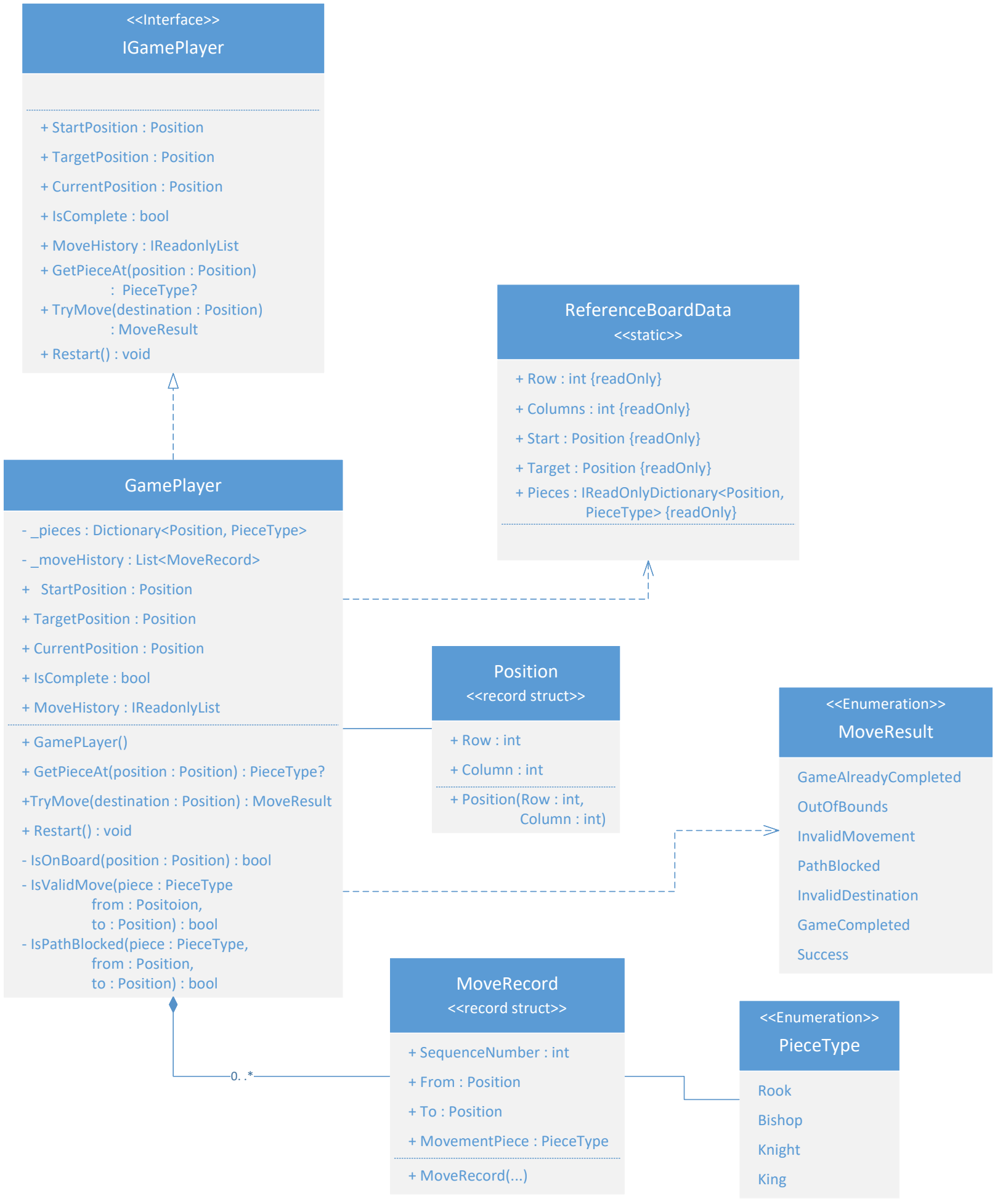

<<Interface>>
IGamePlayer
+ StartPosition : Position
+ TargetPosition : Position
+ CurrentPosition : Position
+ IsComplete : bool
+ MoveHistory : IReadonlyList
+ GetPieceAt(position : Position) : PieceType?
+ TryMove(destination : Position) : MoveResult
+ Restart() : void
ReferenceBoardData
<<static>>
+ Row : int {readOnly}
+ Columns : int {readOnly}
+ Start : Position {readOnly}
+ Target : Position {readOnly}
+ Pieces : IReadOnlyDictionary<Position, PieceType> {readOnly}
GamePlayer
- _pieces : Dictionary<Position, PieceType>
- _moveHistory : List<MoveRecord>
+ StartPosition : Position
+ TargetPosition : Position
+ CurrentPosition : Position
+ IsComplete : bool
+ MoveHistory : IReadonlyList
+ GamePLayer()
+ GetPieceAt(position : Position) : PieceType?
+TryMove(destination : Position) : MoveResult
+ Restart() : void
- IsOnBoard(position : Position) : bool
- IsValidMove(piece : PieceType, from : Positoion, to : Position) : bool
- IsPathBlocked(piece : PieceType, from : Position, to : Position) : bool
Position
<<record struct>>
+ Row : int
+ Column : int
+ Position(Row : int, Column : int)
<<Enumeration>>
MoveResult
GameAlreadyCompleted
OutOfBounds
InvalidMovement
PathBlocked
InvalidDestination
GameCompleted
Success
MoveRecord
<<record struct>>
+ SequenceNumber : int
+ From : Position
+ To : Position
+ MovementPiece : PieceType
+ MoveRecord(...)
0..*
<<Enumeration>>
PieceType
Rook
Bishop
Knight
King